

Exercise #10

20. March 2023

Problem 1.

Compute the Laplace transform of the following functions.

a) $f(t) = 1 + 2t - 4t^4$.

b) $f(t) = 2t^2 e^{3t}$.

c) $f(t) = e^{-2t} \sin(5t)$.

Solution.

- a) We use the Laplace transform $\mathcal{L}(t^n) = \frac{n!}{s^{n+1}}$, and the linearity of the Laplace transform to get

$$\begin{aligned} F(s) &= \mathcal{L}(1) + 2\mathcal{L}(t^2) - 4\mathcal{L}(t^4) \\ &= \frac{1}{s} + \frac{2}{s^2} - \frac{4 \cdot 4!}{s^5}. \end{aligned}$$

- b) We have the s-Shifting property for the Laplace transform, $\mathcal{L}(e^{at}g(t)) = G(s-a)$. If we set $g(t) = 2t^2$ we find the Laplace transformation of $g(t)$ to be $G(s) = \mathcal{L}(2t^2) =$

$$\frac{2 \cdot 2!}{s^3} = \frac{4}{s^3}, \text{ and we obtain}$$

$$F(s) = G(s-2) = \frac{4}{(s-3)^3}.$$

c) Here, we set $g(t) = \sin(5t)$, meaning that $G(s) = \frac{5^2}{s^2 + 5^2}$. We then find the Laplace transform of $f(t)$ to be

$$F(s) = G(s+2) = \frac{5^2}{(s+2)^2 + 5^2}.$$

Problem 2.

Find the inverse Laplace transform of the following functions.

a) $F(s) = \frac{3}{s} + \frac{1}{s^8}.$

b) $F(s) = \frac{2s+1}{s^2-s-2}$

c) $F(s) = \frac{1}{(s+1)(s^2+1)}.$

Solution.

a) We use the following inverse Laplace transform $\mathcal{L}^{-1}\left(\frac{n!}{s^{n+1}}\right) = t^n$, to compute

$$\begin{aligned} f(t) &= 3\mathcal{L}^{-1}\left(\frac{1}{s}\right) + \frac{1}{8!}\mathcal{L}^{-1}\left(\frac{8!}{s^9}\right) \\ &= 3t + \frac{t^4}{8!} \end{aligned}$$

b) Here we first factorize the denominator,

$$F(s) = \frac{2s + 1}{(s - 2)(s + 1)},$$

and then perform a partial fraction decomposition, where,

$$F(s) = \frac{A}{s - 2} + \frac{B}{s + 1}.$$

Then we must have,

$$(s + 1)A + (s - 2)B = 2s + 1.$$

Therefore, we have,

$$(i) \quad A + B = 2$$

$$(ii) \quad A - 2B = 1.$$

We therefore have $(ii) + 2(i) = 3A = 5$, giving, $A = 5/3$. From this we get, $B = 2 - A = 6/3 - 5/3 = 1/3$. This gives,

$$F(s) = \frac{5/3}{s - 2} + \frac{1/3}{s + 1} = \frac{5}{3}\mathcal{L}(e^{2t}) + \frac{1}{3}\mathcal{L}(e^{-t}).$$

Therefore,

$$f(t) = \frac{5}{3}e^{2t} + \frac{1}{3}e^{-t}.$$

c) We first perform a partial fraction decomposition

$$\begin{aligned} \frac{1}{(s + 1)(s^2 + 1)} &= \frac{As + B}{s^2 + 1} + \frac{C}{s + 1} \\ \Rightarrow 1 &= As^2 + Cs^2 + As + Bs + B + C \\ \Rightarrow &\begin{cases} A + C = 0, \\ A + B = 0, \\ B + C = 1 \end{cases} \\ \Rightarrow &\begin{cases} A = -1/2, \\ B = 1/2, \\ C = 1/2 \end{cases} \\ \Rightarrow F(s) &= \frac{1 - s + 1}{2} \frac{1}{s^2 + 1} + \frac{1}{2} \frac{1}{s + 1}. \end{aligned}$$

Then, we find the inverse Laplace transform

$$\begin{aligned} f(t) &= \frac{1}{2} \mathcal{L}^{-1} \left(\frac{1}{s+1} \right) - \frac{1}{2} \mathcal{L}^{-1} \left(\frac{s}{s^2+1} \right) + \frac{1}{2} \mathcal{L}^{-1} \left(\frac{1}{s^2+1} \right) \\ &= \frac{1}{2} e^t - \frac{1}{2} \cos(t) + \frac{1}{2} \sin(t). \end{aligned}$$

Problem 3.

Decide for each of the following statements whether it is true or false. Explain your answer.

- a) If f and g are two functions for which the Laplace transform exists, then $\mathcal{L}(f + g) = \mathcal{L}(f) + \mathcal{L}(g)$.
- b) If f and g are two functions, for which the Laplace transform exists, then $\mathcal{L}(f \cdot g) = \mathcal{L}(f) \cdot \mathcal{L}(g)$.
- c) If the function f satisfies $0 \leq f(t)$ for all $t \geq 0$, then $\mathcal{L}(f)(s) \geq 0$ for all s for which $\mathcal{L}(f)(s)$ exists.
- d) If the function f is continuous and satisfies $0 \leq f(t) \leq 1$ for all $t \geq 0$, then $\mathcal{L}(f)(s)$ exists for all $s > 0$.
- e) If the function f satisfies $0 \leq f(t) \leq 1$ for all $t \geq 0$, then $\mathcal{L}(f)(s)$ exists for all $s > 0$.

Solution.

- a) True. The Laplace transform is a linear transformation (which follows from the fact that integration is a linear operation).
- b) False. We can show an example where this does not hold: We know the Laplace transformation of 1 to be $\frac{1}{s}$. If the statement was true we would have have

$$\frac{1}{s} = \mathcal{L}(1) = \mathcal{L}(1 \cdot 1) = \mathcal{L}(1) \cdot \mathcal{L}(1) = \frac{1}{s} \cdot \frac{1}{s} = \frac{1}{s^2},$$

which is a contradiction.

- c) True. If $f(t)$ is positive, we have that $e^{-st}f(t)$ is positive. Taking the integral of a positive function gives a positive result.
- d) True. $f(t)$ satisfies the growth restriction.
- e) True. $f(t)$ still satisfies the growth restriction.

Problem 4.

Use the Laplace transform in order to solve the following initial value problems:

- a) $2y'' + y' - y = 0, y(0) = 0, y'(0) = 1.$
- b) $y'' + y = 1, y(0) = 0, y'(0) = 0.$
- c) $2y'' + 10y' + 12y = 0, y(0) = -2, y'(0) = 1.$

Solution.

- a) We first do a Laplace transform of the equation, giving,

$$\begin{aligned} 2(s^2 Y(s) - sy(0) - y'(0)) + sY(s) - y(0) - Y(s) &= 0 \\ &= 2s^2 Y(s) + sY(s) - Y(s) = 2. \end{aligned}$$

Therefore,

$$Y(s) = \frac{1}{s^2 + s/2 - 1/2}.$$

We now complete the square in the denominator, giving,

$$Y(s) = \frac{1}{(s + 1/4)^2 - 9/16} = \frac{4}{3} \mathcal{L}(e^{-t/4} \sinh(3t/4)),$$

where we have used that $\mathcal{L}(\sinh(\omega t)) = \omega/(s^2 - \omega^2)$, and the s -shifting rule.

Therefore, $y(t) = \frac{4}{3} e^{-t/4} \sinh(3t/4).$

- b) We Laplace transform the equation, giving,

$$s^2 Y(s) - sy(0) - y'(0) + Y(s) = \frac{1}{s} = s^2 Y(s) + 1 = \frac{1}{s},$$

so that,

$$Y(s) = \frac{1}{s(s^2 + 1)}.$$

If we write,

$$Y(s) = \frac{As + B}{s^2 + 1} + \frac{C}{s},$$

we find that $A = -1$, $B = 0$ and $C = 1$. Therefore,

$$\begin{aligned} y(t) &= -\mathcal{L}^{-1}\left(\frac{s}{s^2 + 1}\right) + \mathcal{L}^{-1}\left(\frac{1}{s}\right) \\ &= -\cos(t) + 1. \end{aligned}$$

c) We first divide by 2 and do a Laplace transformation of the initial value problem

$$\begin{aligned} s^2 Y(s) - sy(0) - y'(0) + 5sY(s) - 5y(0) + 6Y(s) &= 0 \\ \implies (s^2 + 5s + 6)Y(s) + 2s - 1 + 10 &= 0 \\ \implies Y(s) &= \frac{-2s - 9}{(s + 3)(s + 2)} \end{aligned}$$

Next, we can rewrite the first term as follows, Next, we do a partial fraction decomposition to rewrite $Y(s)$

$$\begin{aligned} \frac{-2s - 9}{(s + 3)(s + 2)} &= \frac{A}{s + 3} + \frac{B}{s + 2} \\ \implies -2s - 9 &= A(s + 2) + B(s + 3) \end{aligned}$$

By inserting $s = -2$ and $s = -3$ we find

$$\begin{aligned} A &= 3, \quad B = -5, \\ \implies Y(s) &= \frac{3}{s + 3} - \frac{5}{s + 2}. \end{aligned}$$

Finally, we perform an inverse Laplace transformation to find the solution to the initial value problem

$$y(t) = 3e^{-3t} - 5e^{-2t}.$$

The next exercises are optional and should not be handed in!

Problem 5.

Compute the Laplace transform of the following functions:

a) $f(t) = \sin(4t)e^{3t}$.

b) $f(t) = \sinh(t) \cos(t)$.

Solution.

a) We combine the s-Shifting property and that $\mathcal{L}(\sin(4t)) = \frac{4}{s^2 + 4^2}$, to obtain

$$F(s) = \frac{4}{(s - 3)^2 + 16}$$

b) Rewriting the function gives

$$f(t) = \frac{1}{2} (e^t - e^{-t}) \cos(t).$$

Then, we do a Laplace transform using the s-Shifting property

$$\begin{aligned} F(s) &= \frac{1}{2} \left(\frac{s - 1}{(s - 1)^2 + 1} - \frac{s + 1}{(s + 1)^2 + 1} \right) \\ &= \frac{s^2 - 2}{s^4 + 4} \end{aligned}$$

Problem 6.

Compute the inverse Laplace transform of $F(s)$, where

a) $F(s) = \frac{1}{s^2 + s + 1}$,

b) $F(s) = \frac{as}{s^2 - 2as + a^2 + 1}$ for some $a \in \mathbb{R}$,

c) $F(s) = \frac{1}{s^3 + s}$.

Solution.

a) We rewrite $F(s) = \frac{1}{s^2 + s + 1} = \frac{1}{(s + \frac{1}{2})^2 + \frac{3}{4}} = \left(\frac{3}{4}\right)^{-\frac{1}{2}} \frac{(\frac{3}{4})^{\frac{1}{2}}}{(s + \frac{1}{2})^2 + \frac{3}{4}}$. Now, using the expression for the Laplace transform of $\sin(t)$, we recognize this as $\frac{2}{\sqrt{3}} \mathcal{L}(\sin(\frac{\sqrt{3}}{2}t))$.

b) We rewrite $F(s) = \frac{as}{s^2 - 2as + a^2 + 1} = a \frac{s - a}{(s - a)^2 + 1} + a^2 \frac{1}{(s - a)^2 + 1}$. Using the first shifting theorem, and the expressions for $\mathcal{L}(\sin(t))$ and $\mathcal{L}(\cos(t))$, we recognize this as $F(s) = \mathcal{L}(ae^{at} \cos(t)) + \mathcal{L}(a^2 e^{at} \sin(t))$. Therefore, the inverse becomes $e^{at} a(\cos(t) + a \sin(t))$.

c) We rewrite $F(s) = \frac{1}{s^3 + s} = \frac{1}{s} \frac{1}{s^2 + 1} = \frac{1}{s} \mathcal{L}(\sin(t))$. Now we can use the integration theorem, so that $F(s) = \mathcal{L}(\int_0^t \sin(\tau) d\tau) = \mathcal{L}(1 - \cos(t))$. We therefore get that $f(t) = 1 - \cos(t)$.

Problem 7.

a) Use the Laplace transform to solve the differential equation,

$$y' = -2y + 1, \quad y(0) = 0.$$

b) Use the Laplace transform to solve the differential equation,

$$y''' = -y' + 2, \quad y(0) = y'(0) = y''(0) = 0.$$

Solution.

- a) We compute the Laplace transform of the equation, to get, $sF(s) = -2F(s) + \frac{1}{s}$. Rewritten,

$$F(s) = \frac{1}{s} \frac{1}{s+2} = \mathcal{L}\left(\int_0^t e^{-2\tau} d\tau\right) = -\frac{1}{2}(e^{-2t} - 1).$$

- b) We define $z(t) := y'(t)$, and compute the Laplace transform of the equation, $z^{(2)}(t) = -z(t) + 2$. This gives, $s^2 F(s) = -F(s) + \frac{2}{s}$. Rearranging, we get, $F(s) = 2 \frac{1}{s} \frac{1}{s^2 + 1}$.

We recognize $\frac{1}{s^2 + 1}$ as the Laplace transform of $\sin(t)$, and use the integral theorem to see that $F(s) = 2\mathcal{L}\left(\int_0^t \sin(t)\right) = 2\mathcal{L}(1 - \cos(x))$. Now, we can compute $y(t)$ as $y(t) = 0 + \int_0^t z(u) du = 2t - 2\sin(t)$.